

Gate R2, closed by page-image inspection:
the exact Glimm–Jaffe–Spencer norm, and the corrected constants

Abstract

Gate R2 concerned the prefactor of the GJS norms (1.1.7)–(1.1.8), illegible in the OCR text layer of the JSTOR scan. It is closed by rendering p. 594 of the scan as a 300 dpi image and reading the print directly. The printed norms are

$$\|Q\| = K_1 \left[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)! \right] \|w\|_2, \quad \|Q(x_0)\| = K_1 \left[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)! \right] \|w(x_0, \cdot)\|_2.$$

Two findings. (i) K_1 enters to the *first* power; the exponent ambiguity is resolved as $e_* = 1$. (ii) The local weight is $K_2^{2N(\Delta)}$, *not* $K_2^{N(\Delta)}$: the OCR had dropped the factor 2. Consequently the constant correction “ $K_2^{4p} \rightarrow K_2^{2p}$ ” made in `m2-audit.tex`, Remark 5.3(i), was itself an OCR artifact and is **withdrawn**; the original $K_2^{4p'}$ form of `weak-coupling-strategy.tex` is restored. The cell-count correction $32 \rightarrow 9$ stands (it was geometry, not OCR). All finiteness and uniformity statements of the programme are unaffected — the exponent never entered any qualitative conclusion — but every explicit constant is now exact; the corrected values are propagated through `m6/main.tex` and listed here, including $M^2 = 9 \cdot 8! C_8 K_1 K_2^{16} \lambda^2$ for $\mathcal{P}(\xi) = \xi^4$. The recomputation also caught a second, independent slip in the v1 manuscript — a half-degree error $((2p')!$ for $(4p')!$, $K_2^{4p'}$ for $K_2^{8p'}$) predating R2 — now fixed in `m6/main.tex`.

1 Method and evidence

The JSTOR scan’s embedded text layer garbles the displays (1.1.7)–(1.1.8) of [1] (“ $K_1 [\text{IaK21N(a)} \dots]$ ”), which forced the parametric treatment $K_1^{e_*} \prod K_2^{N(\Delta)} N(\Delta)!$, $e_* \in \{1, r, \sum N(\Delta)\}$, in `m2-audit.tex`. Rendering PDF page 11 (= printed p. 594) at 300 dpi (`pdftoppm`) makes the print unambiguous:

Source Fact 1.1 (p. 594, displays (1.1.7)–(1.1.9), read from the page image).

$$(1.1.7) \quad \|Q\| = K_1 \left[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)! \right] \|w\|_2, \quad (1.1.8) \quad \|Q(x_0)\| = K_1 \left[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)! \right] \|w(x_0, \cdot)\|_2,$$

$$(1.1.9) \quad \|Q\| = \sum_l \|Q_l\| \quad (\text{sum over localized pieces}).$$

Source Fact 1.2 (consistency cross-checks). (a) p. 614 (rendered likewise): the factors $\mathcal{K} = \mathcal{K}_1 \mathcal{K}_2 \mathcal{K}_3 \mathcal{K}_4$ and the category weights K_1^ν , $\nu \in \{1, \dots, 6\}$ (Table 1), belong to the *Chapter-2* norm (2.4.3), not to (1.1.7)–(1.1.8); the Remark preceding Theorem 2.4.1 relates the two by $\|m_\alpha\| = K_1^\nu e^{\tau t} \|m_\alpha\|_0$ for initial and final terms, which is dimensionally consistent only with a *single* power of K_1 inside $\|\cdot\|_0$. (b) $\gamma = (n+1)m_0(1-\varepsilon)$ is printed on p. 614, matching the decay weight used in the imported machinery. (c) Footnote 9 (p. 593) is unchanged: $\kappa = \infty$ is admissible at the price $N(\Delta)! \rightarrow (N(\Delta)!)^3$; no K -factors are affected.

2 Corrected constants

Throughout, $Q = \prod_{i=1}^r \varphi_0(f_i)$ with $\text{supp } f_i$ in a length- L interval, and the localization sum uses at most $L + 1$ cells per factor. Since every point has $m_i = 1$ and $\sum_{\Delta} N(\Delta)$ equals the number of points, the K_2 -weight of any localized piece is $K_2^{2 \cdot \# \text{points}}$ — additive and independent of how points share cells.

Proposition 2.1 (exact norm bounds).

$$\|Q\|_{\text{GJS}} \leq K_1 K_2^{2r} r! \prod_{i=1}^r \sqrt{L+1} \|f_i\|_2, \quad \|Q^2\|_{\text{GJS}} \leq K_1 K_2^{4r} (2r)! \prod_{i=1}^r (L+1) \|f_i\|_2^2,$$

and for the interaction square of (M_2) (two points, degrees $n, n' \leq \deg \mathcal{P} = 2p'$, hence $n + n' \leq 4p'$; ≤ 9 cell pairs):

$$\|Q_V\| \leq 9 \mathfrak{K}(p') \lambda^2 \left(\sum_n |a_n| \right)^2, \quad \mathfrak{K}(p') = K_1 K_2^{8p'} (4p')!.$$

Proof. Immediate from Source Fact 1.1: $\prod_{\Delta} K_2^{2N(\Delta)} = K_2^{2 \sum N(\Delta)}$, $\prod_{\Delta} N(\Delta)! \leq (\sum N(\Delta))!$, Cauchy–Schwarz over $\leq L + 1$ cells per factor, and, for Q_V : $\sum_{\Delta} N(\Delta) = n + n' \leq 4p'$, so $\prod_{\Delta} K_2^{2N(\Delta)} = K_2^{2(n+n')} \leq K_2^{8p'}$ and $\prod_{\Delta} N(\Delta)! \leq (n + n')! \leq (4p')!$, in both cell configurations (shared or distinct squares — the exponent is additive). $\square$

Corollary 2.2 ((M_2) constant, exact form). $\sup_g \|V(h)\Omega_g\|^2 \leq M^2 = 9 C_8 K_1 K_2^{8p'} (4p')! \lambda^2 \left(\sum_{n=0}^{2p'} |a_n| \right)^2$.

For $\mathcal{P}(\xi) = \xi^4$ ($p' = 2$, $\sum |a_n| = 1$; degrees exactly $4+4 = 8$):

$$M^2 = 9 \cdot 8! C_8 K_1 K_2^{16} \lambda^2 = 362\,880 C_8 K_1 K_2^{16} \lambda^2.$$

Corollary 2.3 (exponential-moment radius, slight improvement). $\mathbb{E}_{\mu_g}[\varphi_0(f)^{2n}] \leq C_8 K_1 (K')^{2n} (2n)!$ with $K' = K_2^2 \sqrt{L+1} \|f\|_2$: the prefactor K_1 now sits outside the geometric base, so the radius of exponential integrability $s_0 = 1/K'$ is independent of K_1 — marginally stronger than the v1 statement, where K_1 had to be absorbed into the base to cover all readings.

Remark 2.4 (a correction corrected). `m2-audit.tex`, Remark 5.3(i), “corrected” the constant of `weak-coupling-strategy.tex` from K_2^{4p} to K_2^{2p} . That step trusted the OCR reading $K_2^{N(\Delta)}$ and is hereby withdrawn: with the printed weight $K_2^{2N(\Delta)}$, the *original* $K_2^{4p'}$ was correct. The other half of that remark — the cell count $32 \rightarrow 9$ for supports of length 2 — is geometric and stands. Net constant versus the original strategy claim: $9/32$ of it, with the factorial and K_2 -exponent now tracking $\deg \mathcal{P}$ correctly. (The same recomputation exposed a half-degree slip — $(2p')!$, $K_2^{4p'}$ in place of $(4p')!$, $K_2^{8p'}$ — that had entered `m6/main.tex` at the v0 assembly through a p -versus- $2p'$ convention change between `m2-audit.tex` ($p = \deg \mathcal{P}$) and `m6` ($2p' = \deg \mathcal{P}$); fixed.)

Remark 2.5 (what R2 did and did not affect). No statement of `m6/main.tex` changes its truth value: finiteness and g -uniformity of (M_2) , the gap m_1 in (G) , the rate function Ψ_{m_1} , and all conclusions of the Main Theorem are exponent-insensitive, as asserted throughout. What R2 closes is the *exact form* of every displayed constant; `m6/main.tex` has been updated in place (norm description in §3, Theorem 4.1’s $\mathfrak{K}$, the (G) -section norm bounds, the density-proof radius, the interface checklist). The remaining indeterminacy is irreducible at citation level: GJS fix m_0, K_2, K_1 “sufficiently large” and λ “sufficiently small” without numerical values, so all constants are explicit *in terms of* $(K_1, K_2, C_7, C_8, m_1)$ and cannot be more.

3 Status

- **Closed:** Gate R2, by direct inspection of the page image of p. 594; exponent $e_* = 1$; local weight $K_2^{2N(\Delta)}$; cross-checked against p. 614.
- **Corrected:** the K_2 -exponent correction of `m2-audit.tex` Rem. 5.3(i) is withdrawn; exact constants propagated through `m6/main.tex`; explicit φ^4 value $M^2 = 9 \cdot 8! C_8 K_1 K_2^{16} \lambda^2$; plus the half-degree convention slip in `m6`, found and fixed.
- **No open gates remain** on the weak-coupling closure: R1 was discharged in `r1-check.tex`, R2 here. The Main Theorem of `m6/main.tex` is conditional on the printed theorems of [1] at ordinary citation standard, with all displayed constants in exact form.

References

- [1] J. Glimm, A. Jaffe and T. Spencer, *The Wightman axioms and particle structure in the $\mathcal{P}(\phi)_2$ quantum field model*, Ann. of Math. **100** (1974) 585–632.